

IoT-Based Air Quality Monitoring System with Air Filter

Khukenthra s/o Narayanasamy
Faculty of Electrical and Electronic
Engineering
Universiti Tun Hussein Onn Malaysia
Batu Pahat, Johor, Malaysia
khukenn@gmail.com

Sharifah Binti Saon
Faculty of Electrical and Electronic
Engineering
Universiti Tun Hussein Onn Malaysia
Batu Pahat, Johor, Malaysia
sharifa@uthm.edu.my

Abd Kadir Mahamad
Faculty of Electrical and Electronic
Engineering
Universiti Tun Hussein Onn Malaysia
Batu Pahat, Johor, Malaysia
kadir@uthm.edu.my

Maisara Binti Othman
Faculty of Electrical and Electronic
Engineering
Universiti Tun Hussein Onn Malaysia
Batu Pahat, Johor, Malaysia
sharifa@uthm.edu.my

Abstract— Due to industrialization and globalization, which have damaged natural resources, technologies to detect and monitor contaminants in the air have been made possible via the Internet of Things (IoT). It is important to monitor the quality of air to conserve the quality of the surrounding air. An IoT-based system helps monitor and collect real-time data about air contaminants such as carbon monoxide (CO), carbon dioxide (CO₂), Liquefied Petroleum Gas (LPG), Methane (CH₄), benzene, and nitrogen oxides (NO_x). Air quality is a national issue that should be taken care of to prevent any major health complications to any live forms. Poor air quality causes serious health concerns, including respiratory and cardiovascular disorders, allergies, and early mortality. The system utilizes a range of sensors and real-time data analysis techniques to monitor indoor air quality and activate AC fans when pollutant levels exceed predefined thresholds. The data is accessible via the ThingSpeak platform and a custom Android application, providing users with timely alerts through multiple communication channels. The results demonstrate the system's effectiveness in maintaining air quality and its potential health benefits.

Keywords—IoT technology, air quality monitoring system, ThingSpeak

I. INTRODUCTION

Air is a consistent pressure that cannot be seen by the naked eye but can be felt. The swinging of trees and plants can be taken as a primary example of how air may be seen to have an impact on nature. It is known for certain that life does not exist without the presence of air, which means that without air, the Earth would be just identical to the other lifeless planets in our solar system. Apart from that, if air didn't exist, there wouldn't have been any sound, and humans would be encased in a vacuum bubble because, without air, sound cannot travel. The composition of air can be divided into several parts, which are 78% nitrogen (N₂), 20% oxygen (O₂), 0.04% carbon dioxide (CO₂), 0.93% argon (Ar), 0.0018% neon (Ne), 0.000524% helium (He), 0.000187% methane (CH₄) and 1% of other gases [1]. Therefore, the concentration of N₂ in the air is more than the concentration of O₂ in the air. Although fresh air can be obtained from outside, it is also important to maintain good and fresh air indoors because most of the people nowadays are working in a closed environment and spending less time outside. So, in this study, an air quality monitoring device has been designed to check the quality of air indoors.

This work was sponsored by Universiti Tun Hussein Onn Malaysia.

The world is now moving towards modern technologies. A lot of new machines and gadgets were invented to make the job much easier. From a little tailor machine to the fastest car in the world, from wired telephones to mobile smartphones, a lot of evolution has gone through and is still ongoing. To achieve this, a lot of factories, buildings, and companies have been constructed. Moreover, the world population has also increased from 7 billion to 8 billion lives [2]. Therefore, a lot of deforestation occurs to construct houses for accommodation purposes and some residential areas were constructed near the factories. Although it is good to have modern technologies and an easy lifestyle, this rapid growth of modern technologies brings along environmental crisis and one of the examples is air pollution.

The smoke produced by the factories and companies consists of a lot of contaminants such as hydrogen chloride, dioxin, benzene, or toluene, compounds like asbestos, or elements such as chromium, mercury, and cadmium. If there are houses near this kind of factory, it is more likely for this smoke to enter the house and may affect the daily intake of air by the residents. Apart from that, smoking habits are now becoming a common practice among people nowadays. Cigarette smoke contains nicotine, carbon monoxide, tar, and toxic chemicals such as benzene, arsenic, and formaldehyde. Even though people know the harmful effects of smoking, still it has been practiced by a lot of people. This is not just affecting the health of smokers, but it can also affect the people around these smokers known as passive smokers or second-hand smokers. When a non-smoker is beside an active smoker, the non-smoker tends to inhale the smoke from the active smoker without knowledge. On the other hand, compared to active smokers, passive smokers are exposed to a different kind of smoke [3]. So, if a person in a house smokes, then it may affect the other family members in that house including children. Furthermore, open burning is also carried out in a lot of residential areas even though it is illegal. A lot of people can be a victim of the smoke from this open burning. It can be more dangerous since anything can be burned such as plastic materials like plastic bags, plastic bottles, dry leaves, and clothes. Therefore, the smoke emitted from this kind of open burning may consist of nitrogen oxides, sulfur dioxide, volatile organic chemicals (VOCs), and polycyclic organic matter (POMs) which is bad for health [4].

Natural (NA) gas and liquefied petroleum gas (LPG) are the two main gases used for cooking at home. LPG is a flammable and colorless gas. LPG is a mixture of 48%

propane, 50% butane and 2% pentane [5]. It is widely accessible and kept in pressurized tanks in most places. The permitted level for butane is 600 ppm above which is not safe [6]. In addition, it serves as a refrigerant, lighter fuel, aerosol propellant, petrol replacement, and input into the synthesis of other chemicals and polymers. Moreover, LPG is an asphyxiant gas that, if oxygen levels are sufficiently lowered, can result in unconsciousness and/or death. In addition to being a gas under pressure, LPG is also highly combustible and can explode when heated. The major ways that people are exposed to LPG are by inhalation, skin contact, and eyes. Even brief exposure to high concentrations of LPG vapor in the air can result in asphyxiation, convulsions, comas, cardiac issues, and even death. Prolonged exposure can cause weariness, weight loss, conjunctivitis, nosebleeds, rhinitis, oral/nasal ulcerations, and harm to the central nervous system [7].

On the other hand, natural gas stoves release methane (CH_4) which is a potent greenhouse gas, and other pollutants through leaks and incomplete combustion. Methane escaping from stoves inside American homes is thought to have an equivalent climatic impact as around 500,000 gasoline-powered motor vehicles, according to Stanford researchers [8]. Two ways that natural gas appliances contribute to global warming are by releasing unburned methane into the atmosphere and by producing carbon dioxide by burning natural gas as fuel. It was discovered that when stovetops were not in use, more than 75% of the methane emissions from stovetops leaked into homes [9]. Naturally occurring in large quantities, methane is the primary component of natural gas, which is utilized extensively in homes and businesses and may be fatal to humans if inhaled in high enough proportions. According to these results, gas stoves have a greater environmental effect even if they could be more convenient for cooking.

II. METHODOLOGY

A. Block Diagram

In the proposed system, NodeMCU ESP8266 with Wi-Fi functionality has been used, which is suitable for this project. For the activation of this entire system, a power supply is needed. Once the gas sensors MQ135, MQ4, MQ7, MQ2, and the temperature and humidity sensor DHT11 are activated, the data can be monitored on the liquid crystal display (LCD) screen. Each gas sensor, MQ135; MQ7; MQ4; and MQ2; can detect and measure carbon monoxide (CO), nitrogen oxides (NO_x), benzene, and carbon dioxide (CO_2); natural gas and methane (CH_4); and smoke and liquefied petroleum gas (LPG).

Meanwhile, if the quality of air exceeds the limit of ppm that has been set for each sensor, all AC fans are activated. The data is recorded and can be monitored through the mobile application which has been developed. The block diagram of this project is presented in Fig. 1.

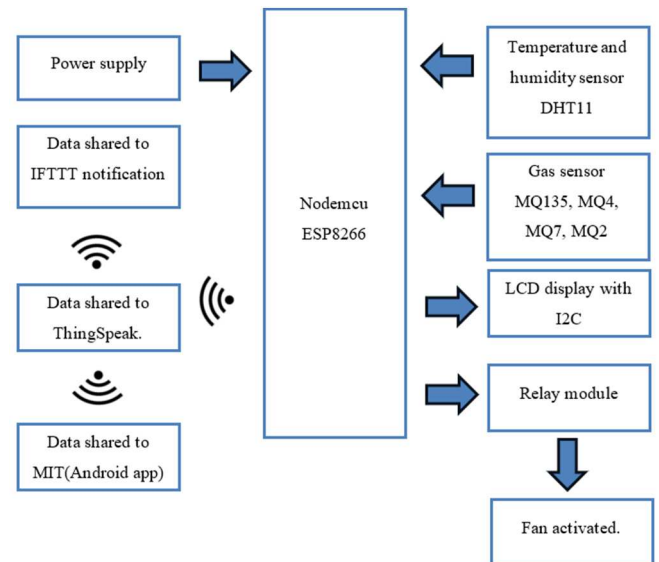


Fig. 1. Project block diagram

B. Flowchart

Fig. 2 shows an overview of the program flowchart of the system. It shows the flow of the project of five AC fans get triggered based on the gas sensors' readings. All the five AC fans are turn on in parallel even if only one sensor reading exceeds the threshold value. While, the AC fans are turn off, once the readings drop below the threshold value.

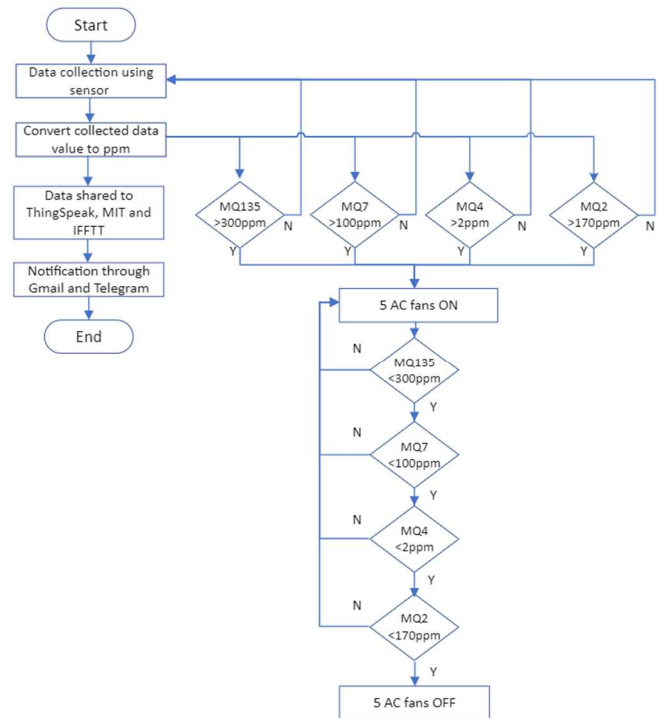


Fig. 2. Project Flowchart

III. RESULTS AND DISCUSSION

A. Model design

As per Fig. 3 and Fig.4, the model with dimensions of 40cm x 40cm x 40cm, consists of five AC fans which are activated once the readings of either one gas sensor exceed the ppm level. As in Fig.3, two AC fans are for exhaust purposes, and in Fig. 4, three AC fans are for vacuum purposes where one vacuum fan, that placed at the top of the model. These AC fans are connected to the 240V supply and NodeMCU

ESP8266 is connected to the 5V power supply. High-efficiency particulate (HEPA) air filters; 26.5cm x 34cm, have been used to filter the contaminants of the air. The sensors have been placed in an upright position so that the smoke can reach the sensors easily.



Fig. 3. Back view of the model



Fig. 4. Front view of the model

B. Data Analysis

As in Fig. 5, Table I, Fig. 6, and Table II, the data of the sensors were obtained in the ThingSpeak platform. In Fig. 5 and Table I, the data shows the values of sensors before calibrating the sensors which are in ThingSpeak and Excel respectively, and in Fig. 6 and Table II, the data shows the values of sensors after calibrating the sensors. For the calibration of sensors, the formula used was $y = mx + c$ where m is the slope, c is the intercept, x is the analog value of the sensors and y is the converted value (ppm). First, the data were collected to monitor the sensors' sensitivity towards normal environments and bad environments. Incense sticks were used to create a bad environment since smoke produced from incense sticks contains common VOCs including benzene, toluene, xylenes, and isoprene. MQ135 and MQ7 were used to detect smoke. To create a gaseous environment, a lighter was used which contains a reservoir of lighter fluid, a liquefied flammable gas such as propane or butane, and cooking gas which releases methane. MQ4 and MQ2 were used to detect these gases. The sensors were placed in an enclosed room with a dimension of 2.5m x 3m with no fan for around 12 hours to obtain the data to check the values in a normal environment for calibration. Then, the average of analog values was taken

to calibrate each sensor. It can be seen in Fig. 5, that the data is not stable and fluctuates without the presence of any smoke. In Fig. 6, in the Field 3 chart of MQ4, the value is in analog value to show that the analog value obtained throughout that time for all the gas sensors was stable due to the calibration process.



Fig. 5. The results of MQ7, MQ135, MQ4, MQ2, DHT11 (temperature), DHT11 (humidity) before calibration in ThingSpeak.

TABLE I. THE RESULTS OF MQ7, MQ135, MQ4, MQ2, DHT11 (TEMPERATURE), DHT11 (HUMIDITY)

	A	C	D	E	F	G	H
390	created_at	field1	field2	field3	field4	field5	field6
391	2023-12-14T18:01:30+08:00	1138	307	7918	29.8	88	3386
392	2023-12-14T18:02:18+08:00	1045	297	7747	29.8	88	7477
393	2023-12-14T18:06:42+08:00	818	277	7309	29.8	88	3396
394	2023-12-14T18:08:41+08:00	775	272	8145	29.8	88	2929
395	2023-12-14T18:09:54+08:00	755	267	8710	29.8	88	2747
396	2023-12-14T18:11:30+08:00	732	264	7873	29.8	88	2588
397	2023-12-14T18:11:54+08:00	732	261	7780	29.8	87	2543
398	2023-12-14T18:16:18+08:00	686	256	7227	29.4	87	2263
399	2023-12-14T18:20:41+08:00	671	252	6997	29.8	87	2080
400	2023-12-14T18:21:53+08:00	667	249	6968	29.8	87	2038
401	2023-12-14T18:22:42+08:00	662	248	6929	29.3	87	1998
402	2023-12-14T18:27:55+08:00	13136	272	7192	29.8	87	2162
403	2023-12-14T18:29:07+08:00	1862	268	7137	29.8	87	2025
404	2023-12-14T18:46:59+08:00	764	278	7330	30.2	85	2186
405	2023-12-14T18:49:03+08:00	600	270	7154	29.8	85	1976
406	2023-12-14T18:49:27+08:00	16496	274	7168	29.8	85	2035
407	2023-12-14T18:50:39+08:00	896	274	7115	29.8	85	5835
408	2023-12-14T18:51:03+08:00	750	266	7080	29.8	86	6826
409	2023-12-14T18:53:27+08:00	556	264	6965	29.3	86	6168
410	2023-12-14T18:59:27+08:00	514	266	6910	29.8	86	3162
411	2023-12-14T19:01:27+08:00	485	262	6846	29.8	86	2738

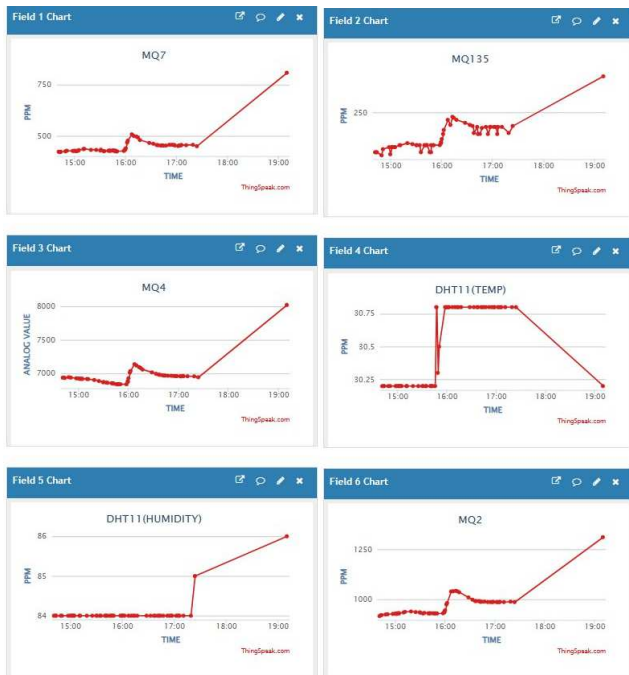


Fig. 6. The results of MQ7, MQ135, MQ4, MQ2, DHT11 (temperature), DHT11 (humidity) after calibration in ThingSpeak.

TABLE II. THE RESULTS OF MQ7, MQ135, MQ4, MQ2, DHT11(TEMPERATURE), DHT11(HUMIDITY)

	A	C	D	E	F	G	H
126	created_at	field1	field2	field3	field4	field5	field6
127	2023-12-16T15:01:11+08:00	429	216	6925	30.2	84	929
128	2023-12-16T15:01:59+08:00	427	216	6920	30.2	84	928
129	2023-12-16T15:02:48+08:00	427	216	6924	30.2	84	931
130	2023-12-16T15:04:24+08:00	429	216	6921	30.2	84	930
131	2023-12-16T15:04:48+08:00	432	216	6922	30.2	84	932
132	2023-12-16T15:10:23+08:00	437	218	6920	30.2	84	935
133	2023-12-16T15:11:36+08:00	438	218	6918	30.2	84	938
134	2023-12-16T15:19:11+08:00	433	220	6905	30.2	84	940
135	2023-12-16T15:25:10+08:00	433	219	6889	30.2	84	937
136	2023-12-16T15:29:58+08:00	430	218	6872	30.2	84	936
137	2023-12-16T15:30:46+08:00	434	218	6874	30.2	84	934
138	2023-12-16T15:33:58+08:00	427	218	6866	30.2	84	930
139	2023-12-16T15:35:10+08:00	427	211	6863	30.2	84	933
140	2023-12-16T15:39:58+08:00	429	218	6857	30.2	84	931
141	2023-12-16T15:41:10+08:00	430	218	6853	30.2	84	931
142	2023-12-16T15:41:34+08:00	428	218	6855	30.2	84	931
143	2023-12-16T15:42:46+08:00	429	218	6850	30.2	84	931
144	2023-12-16T15:45:34+08:00	431	211	6842	30.2	84	931
145	2023-12-16T15:47:10+08:00	427	211	6844	30.8	84	930
146	2023-12-16T15:47:35+08:00	430	218	6840	30.8	84	930
147	2023-12-16T15:49:11+08:00	425	218	6842	30.3	84	931

C. Android Application

As in Fig. 7, the application interface was created using the MIT app inventor. It is a simple Android app that shows the results of each sensor obtained through the ThingSpeak platform. The purpose of this Android app is to monitor or access the sensors' data anytime in any place easily.

When the user clicks "Enter", the sensors screen as in Fig. 7 (a) can be seen. The user can select any icon to view the data of a particular sensor. For example, if the smoke icon is clicked as in Fig. 7 (b), while the results screen as in Fig. 7 (c) is displayed.

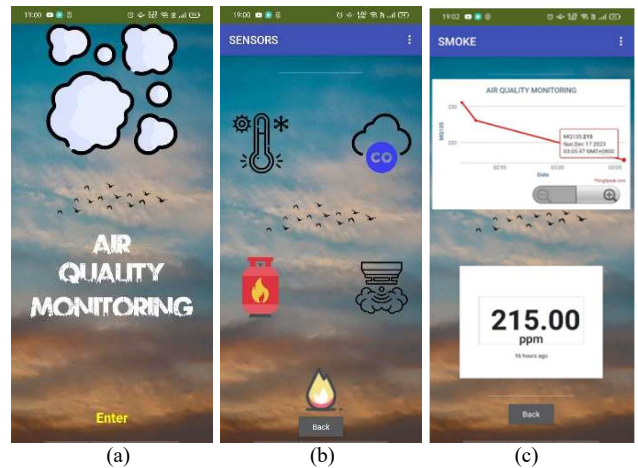


Fig. 7. Android application (a) main screen, (b) selection screen, and (c) results screen of the MQ135 sensor.

D. Notification

As in Fig. 8, alerts are received in Gmail, phone messages, and Telegram respectively once any gas sensor detects bad air quality.

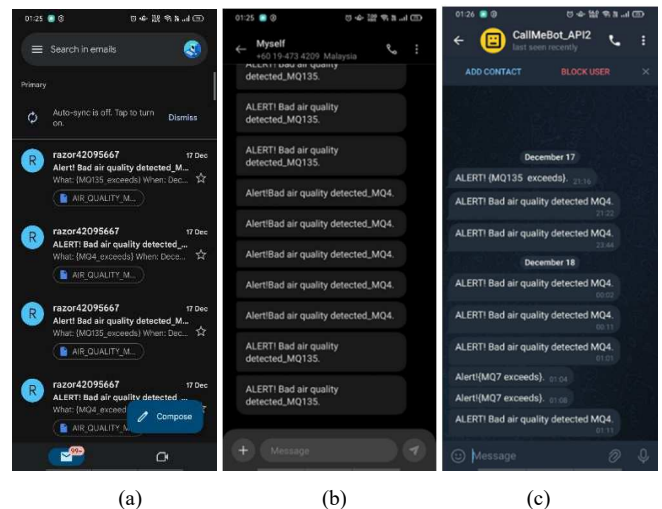


Fig. 8. Notification (a) inbox of Gmail, (b) inbox in phone, and (c) inbox in telegram.

E. Verification of data

As per the findings, the project was tested in a closed room with no ventilation to ensure that the readings obtained were not affected by other gases and to stabilize its readings and ensure accurate measurements. It can be seen that the projected values in ThingSpeak as shown in Fig. 5, were not steady and fluctuating and it was before calibration. To calibrate the sensors, the actual ppm limit was referred to some journals and other similar projects. According to [10], there is around 400 ppm of carbon dioxide in the atmosphere, the average indoor values range from 35-450 ppm and the maximum acceptable level is around 1000 ppm. The MQ-135 detected 426 ppm inside of a room and up to 1200 ppm of carbon dioxide as a match burned under it before it was extinguished. Another source states that the safe level of carbon dioxide is 400-1,000 ppm concentrations typical of occupied indoor spaces with good air exchange [11]. So, to calibrate the value of the MQ135 sensor, the threshold value has been set to 300 ppm. This is because, although the safe level ranges between 400-1000ppm, according to [12], a good argument could be made that 280 ppm is the ideal level of carbon dioxide for human life since it creates temperature

ranges that are comfortable for the human body and allowed civilization to grow.

For the MQ7 sensor, the carbon monoxide exposure range has been divided into four levels which are low-level, mid-level, high-level, and dangerous level. 50 ppm and less is considered a low level, between 51 ppm and 100 ppm is considered a mid-level, greater than 101 ppm if no one is experiencing symptoms is considered a high level, and greater than 101 ppm if someone is experiencing symptoms is considered as dangerous level [13]. Moreover, 0-50 ppm can be considered completely safe, 100-150 ppm can be considered unhealthy, but only for sensitive groups, and above 151 ppm is completely unsafe or unhealthy.

While, MQ4 able to detect methane level in the air, with normal methane level is 1.7 ppm. Once the methane level increase, bad air level is considered, and all AC fans activated.

IV. CONCLUSION

The goal of the air quality monitoring device is to measure the standard of the indoor air that everyone breathes daily. Nowadays, practically anything can be done online from home using a variety of devices, including mobile phones and laptops. Additionally, several online resources make life easier for everyone, including online versions of newspapers, magazines, and television websites. People frequently work from home as a result of saving time and energy. This causes people to spend a lot of time indoors, either directly or indirectly. Additionally, individuals frequently burn illegally in the open and smoke inside their homes, both of which have an impact on the air quality. Moreover, a lot of people used to cook in their homes using natural gas or Liquefied Petroleum Gas (LPG) which may cause gas leakage, which is not good for health and can cause fire accidents. So, it's important to check whether the interior air quality is good or not. Thus, this project can both assist in maintaining high air quality and assist individuals in ensuring that the air they are breathing is safe.

ACKNOWLEDGEMENT

This work was sponsored by Universiti Tun Hussein Onn Malaysia.

REFERENCES

- [1] Helmenstine, A. (2022). Chemical Composition of Air. Science Notes and Projects. https://sciencenotes.org/chemical-composition-air/#google_vignette
- [2] Haymond, D. W. R. M. L. S. H. (2022). The world population hit 8 billion. Here's where you fit in. Washington Post. <https://www.washingtonpost.com/world/interactive/2022/world-population-8-billion/>
- [3] Sébastien Schramm, Vincent Carré, Jean-Luc Scheffler, Frédéric Aubriet, Active and passive smoking – New insights on the molecular composition of different cigarette smoke aerosols by LDI-FTICRMS, *Atmospheric Environment*, Vol. 92, 2014, pg. 411 - 420, <https://doi.org/10.1016/j.atmosenv.2014.04.052>.
- [4] Environmental and health impacts of open burning | Wisconsin Department of Natural Resources, <https://dnr.wisconsin.gov/topic/OpenBurning/Impacts.html>
- [5] M. H. B. M. Yaya, R. K. Patchmuthu and A. T. Wan (2021). LPG Gas Usage and Leakage Detection Using IoT in Brunei. 2021 International Conference on Green Energy, Computing and Sustainable Technology (GECOST), Malaysia, pp. 1-5. doi: 10.1109/GECOST52368.2021.9538647.
- [6] R. K. Kodali, T. Devi B. and S. C. Rajanarayanan (2019). IoT Based Automatic LPG Gas Booking And Leakage Detection System. 2019 11th International Conference on Advanced Computing (ICoAC), Chennai, India, pp. 338-341. doi: 10.1109/ICoAC48765.2019.246863.
- [7] Ahmadi, O., Mortazavi, S.B., & Mahabadi, H.A. (2020). Review of atmospheric storage tank fire scenarios: costs and causes. *Journal of Failure Analysis and Prevention*, 20, 384–405. doi:10.1007/s11668-020-00987-6.
- [8] Ranjan, R., Singh, S. Switching Towards LPG: Indian Household Perspectives. *J. Quant. Econ.* 21, 417–435 (2023). <https://doi.org/10.1007/s40953-023-00337-y>.
- [9] Reddy, B.V., Boorla, N., Kakarlapudi, B.S., Nadirge, D.S., & Siripangi, B. (2023). IoT based air quality monitoring system. *AIP Conference Proceedings*, 2794, 020009. doi:10.1063/5.0173454
- [10] Wang, R.; Li, W.; Gao, J.; Zhao, C.; Zhang, J.; Bie, Q.; Zhang, M.; Chen, X. The Influence of Bedroom CO₂ Concentration on Sleep Quality. *Buildings* 2023, 13, 2768. <https://doi.org/10.3390/buildings13112768>.
- [11] Marques, G., Ferreira, C.R. & Pitarma, R. Indoor Air Quality Assessment Using a CO₂ Monitoring System Based on Internet of Things. *J Med Syst* 43, 67 (2019). <https://doi.org/10.1007/s10916-019-1184-x>
- [12] Nunes, L.J.R. The Rising Threat of Atmospheric CO₂: A Review on the Causes, Impacts, and Mitigation Strategies. *Environments* 2023, 10, 66. <https://doi.org/10.3390/environments10040066>
- [13] Naghizadeh, A., Sharifzadeh, G., Tabatabaei, F. et al. Assessment of carbon monoxide concentration in indoor/outdoor air of Sarayan city, Khorasan Province of Iran. *Environ Geochem Health* 41, 1875–1880 (2019). <https://doi.org/10.1007/s10653-018-0226-5>